Benedict Armstrong

Curriculum Vitae

Computational Science and Engineering, ETH Zürich

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Computational Science and Engineering student at ETH Zürich working on robot learning and at the intersection of machine learning and physics. At the Robotic Systems Lab I work on world models for contact-rich manipulation, using differentiable simulation with smoothed contact, and at the ETH Robotics Club I train whole-body control policies for a humanoid robot. Before that, I developed state-space sequence models for low-latency detection of gravitational-wave signals. I also have several years of experience shipping full-stack software as a freelancer and have taught courses on algorithms, data structures and software engineering.

1 Education



Figure 1: Born and raised in Vienna, currently living in Zürich. In my free time, I like to code, tinker with electronics, design and build furniture, and generally enjoy all types of DIY/maker projects. I am also interested in robotics, drones, IoT prototyping, 3D printing, tennis, cycling, triathlons, and of course, all things computer science.

M.Sc. Computational Science and Engineering
ETH Zürich — 2026 – present
Robotics track

B.Sc. Computational Science and Engineering
ETH Zürich — 2021 – 2025
Robotics track

B.Sc. Computer Science
Vienna University of Technology — 2019 – 2021 (switch to ETH)
Software & Information Engineering track

2 Research

World Models for Contact-Rich Robotics
Semester Project and Seminar, Robotic Systems Lab (RSL), ETH Zürich — Fall 2026 – present

Joint semester project and *CSE Seminar in Robotics* with Tristan Gabl on world models for contact-rich robot manipulation, with robot cloth folding as the target task.

- Surveyed latent, video, particle-based and differentiable-physics world models and their evidence on contact-rich tasks such as insertion and cloth folding.
- Investigating soft contact gradients (smoothing, straight-through estimation, randomized smoothing) to make differentiable simulators in MJX usable as world models corrected from real data.

Humanoid Robot Learning
Humanoid Team, ETH Robotics Club — Fall 2026 – present

Robot learning for the ETH Robotics Club's humanoid robot: motion retargeting and learned whole-body control, from training in simulation to deployment on the robot.

- Trained a whole-body teleoperation and walking policy for the club's V1 humanoid (22 DoF) with InstantTrack in about one hour on a single GPU.
- Made the policy robust to the real robot's IMU error and command latency, and validated it sim-to-sim against the robot's own drive and sensing models.
- Online demo: the trained policy walks the humanoid in MuJoCo in the browser, controlled with the keyboard or touch.

AI for Gravitational-Wave Physics
ML Research Intern, Computational Applied Mathematics & AI Lab (CAMAIL)
June 2025 – Summer 2026

ELLIS Institute Tübingen and Max Planck Institute for Intelligent Systems, in collaboration with the LIGO Lab at MIT supervised by T. Konstantin Rusch.

Gravitational-wave signals can last minutes and are buried in non-stationary, glitchy detector noise, while

astronomers need an alert within seconds to point telescopes at the merger. I worked on Linear Oscillatory State-Space Models (LinOSS): sequence models that handle very long inputs efficiently and towards making it possible to detect binary neutron star mergers in real time.

- Designed and trained state-space models for low-latency binary neutron star detection on LIGO strain data.
- Built the training and evaluation pipeline in PyTorch (Lightning) and JAX/Equinox.
- Core contributor to Discretax, the lab's open-source library of state-space models in JAX.
- Co-developed PhyTS, a benchmark of physics time-series tasks (NeurIPS 2026).

Results appeared at NeurIPS 2026 [1] and were presented at AI for Gravitational Waves Workshop (CERN, Geneva) [2] and Fast Machine Learning for Science Conference 2026 (UC San Diego) [3].

3 Publications and Talks

[1] **Benedict Armstrong**, Jeroen Audenaert, Hannah P. Binney, Alice Cheng, Maarten De Vos, Luis F. Domingues, et al. PhyTS: A Benchmark for Scientific Time Series. In *Advances in Neural Information Processing Systems (NeurIPS)*, Sydney, 2026.

[2] **Benedict Armstrong**, Bhavya Gupta, Christina Reissel, Erik Katsavounidis, Kyungseop Yoon, Philip Harris, et al. Linear Oscillatory State-Space Models for Binary Neutron Star Detection. Talk at *AI for Gravitational Waves Workshop*, CERN, Geneva, May 2026.

[3] Kyungseop Yoon, **Benedict Armstrong**, Christina Reissel and Philip Harris. Direct Parameter Estimation of Binary Neutron Star Mergers with State Space Models. Talk at *Fast Machine Learning for Science Conference 2026*, UC San Diego, September 2026, presented by Kevin (Kyungseop) Yoon.

4 Experience

Freelance Software Engineer

Armstrong Systems — 2022 – 2024

Founded Armstrong Systems, software consulting for clients in Zürich. Full-stack development of various projects including a web-based data management tool, visualization dashboards and large-scale data ingestion pipelines.

- Scoped, built and maintained a web-based data management tool used day to day by the client's team.
- Built interactive visualization dashboards on top of the clients' operational data.
- Designed large-scale data ingestion pipelines, from source integration to deployment and monitoring.

Coding Assistant for LehrFEM++ Finite Element Library

Seminar for Applied Mathematics, ETH Zürich — Fall 2024

Coding assistant for the C++ finite element library LehrFEM++, used in the course *Numerical Methods for Partial Differential Equations* taught by Prof. Dr. Ralf Hiptmair.

- Extended and maintained the LehrFEM++ codebase used by students in the course's programming exercises.
- Wrote documentation and examples to make the library easier for students to pick up.

Paramedic (civil service)

Samariterbund — 2018 – 2019

Worked as a paramedic in Vienna as part of my civil service.

Various summer jobs

Various — Summers 2017–2021

Various summer jobs in Vienna, including working as a postman, food delivery driver and math/CS tutor for high school students.

5 Teaching

Table 1: Teaching positions, grouped by institution.

Role	Course	Period
Department of Computer Science, ETH Zürich		
Teaching Assistant	*Software Engineering Fundamentals*	Spring Semester 2024
Head Teaching Assistant	*Computer Science II*	August 2023 – February 2024
Teaching Assistant	*Computer Science II*	August 2022 – February 2023
Research Group Industrial Software (INSO), TU Wien		
Teaching Assistant and Full Stack Developer	*Security for Systems Engineering*	Spring 2021

6 Projects

Presio: presio.ch · GitHub
Web app for presenting PDF slides with a presenter controller and a mirrored audience window. PDFs stay in the browser by default; signed-in users can share a session code so others can follow live on their own devices. Since its launch in September 2026 it has had over 2,000 users.

AcceptOdds: acceptodds.com · GitHub
A prediction market for research papers, launched on the ICLR 2027 submissions. Researchers with an institutional email trade each paper's outcome (Accept/Reject) with a non-convertible play currency. Prices come from a logarithmic market scoring rule (LMSR). Since launching at the end of September 2026 it has had about 54,000 unique visitors from 96 countries and over 600,000 page views.